

FFGFT — Epistemological Position

4D Description, Natural Units, and Self-Referentiality

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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Abstract

This document summarises the epistemological position of the Fundamental Fractal Geometric Field Theory (FFGFT) as it has developed through ongoing discussions within the Information Physics Institute (IPI). It addresses: (1) the status of the 4D description (fractal, open, recursive), (2) the equivalence of all descriptions in natural units, (3) the structure of the vacuum and the T-field, (4) the FFGFT position on black holes, and (5) the unavoidable self-referentiality of any description of a physical system from within. The document serves as a foundation for the continuing IPI discussion on the question of primitivity (geometry vs. information).

Contents

.1 The 4D Description Is Fractal, Open, and Recursive

FFGFT works with the 4D identification torus structure T^4 — but this description is not a closed, deterministic system. It is:

Fractal: The same geometric structure (ξ -ladder, winding numbers) appears at different scales. The fractal dimension $D_f = 2.999867$ is extremely close to 3 — the deviation is subtle but physically consequential.

Open: The first step — why T^4 , why $\xi = 1.333 \times 10^{-4}$ — remains explicitly open. FFGFT makes no ontological completeness claim.

Recursive: The ξ -ladder is a recursive structure with a natural cut-off at $N \approx 8-9$, where the Z_3^3 topological structure and Fibonacci convergence coincide.

.2 The 4D Description Does Not Apply to the Totality

FFGFT does *not* claim that the entire domain of existence is described by 4D and nothing further exists beyond it. The 4D description is a framework for what is physically meaningful within its bounds — not a statement about everything that could or could not exist.

FFGFT is not metaphysical absolutism. It is a derivation framework: *given T^4 and ξ , everything else follows*. What lies before T^4 remains open — a deliberate, honest choice.

.3 The Lower Bound: 4D Applies to the Absolutely Smallest

The minimal length scale:

$$L_0 = \xi \cdot \ell_P \approx 2.2 \times 10^{-39} \text{ m} \quad (1)$$

is the lower boundary of the 4D framework. Below L_0 , T^4 has no well-defined degrees of freedom.

This boundary applies *downward*, not upward. There is no upper bound in the sense of a maximum scale of the universe — torus size is an emergent coherence length, not a fixed parameter. The question of the “total size” of T^4 is structurally ill-posed.

.4 Natural Units: No Privileged Description

In natural units, all fundamental descriptions are equivalent [?]:

$$E \leftrightarrow m \leftrightarrow \frac{1}{t} \leftrightarrow f \quad (2)$$

with $E = mc^2$, $E = hf$, $\tilde{T} \cdot m = 1$ (FFGFT's foundational relation), and $f = 1/t$.

None of these descriptions is privileged. They are different projections of the same geometric reality on T^4 . The choice of representation is a matter of convenience, not ontological priority.

.5 Consequence for the Question of Information vs. Geometry

“Information” as a description is a valid projection — but not more fundamental than the geometric description. It is equivalent, not primary.

Information Is Distinction

Information is distinction. Shannon1948 (Shannon1948), Bateson1972 (Bateson1972), and the IPI discussions of 2026 [?] all converge on the same point: without distinction, there is no information. "Pure unrestrained information" without structure is indistinguishable from nothing — it is not a primitive but the absence of everything.

The decisive point: *unstructured* information is not an equivalent projection of T^4 . It is a postulated state *prior to* all structure — and therefore outside the domain about which FFGFT makes claims.

[Consequence for Atticus's Postulate] Atticus Myser postulates "pure unrestrained information" as primitive — without structure, without distinction, without binding. But information without distinction is no longer an information concept. It is an empty name. What he describes — continuous spin, unbounded velocity, no quantum numbers — is not "pure information" but **the absence of all structure through which information would be defined**. Such a postulate cannot generate field equations, cannot span a torus, cannot explain occlusion. It is not wrong — it is **not connectable** to any known or coherently conceivable physics.

.6 The Vacuum Has Structure — No Field Without Structure

The Vacuum Is Not Empty

In FFGFT, the vacuum is the ξ -field ground state — not empty space but a geometric minimal structure with well-defined properties:

- Vacuum energy density as a geometric property of the ξ -field
- T-field configuration $\tilde{T} \cdot m = 1$ active even in the ground state
- Minimal length scale $L_0 = \xi \cdot \ell_P$ as a structural lower bound

Quantum field theory likewise knows no structureless vacuum — zero-point fluctuations, vacuum polarisation, and the Casimir effect all demonstrate: the vacuum *is* structure.

Thermodynamic Equilibrium Changes Nothing

Even if the vacuum or the field were in total thermodynamic equilibrium — maximum entropy, all fluctuations averaged out — the field would remain structured:

- Field equations still hold
- T^4 topology still exists
- ξ retains its value

Important: Equilibrium is not the absence of structure. It is a specific, distinguished state within the structure. Equilibrium presupposes a space of structure within which equilibrium is *definable*. Without field structure, there is no concept of equilibrium.

FFGFT Is a Field Theory — No Field Without Structure

A field is mathematically a mapping from points of a carrier space to values obeying field equations. This necessarily requires:

- A carrier space (structure)
- A mapping rule (structure)
- Field equations as constraints (structure)

No field without structure — this is a mathematical tautology. “Pure unstructured information” cannot carry a field, cannot build a vacuum, cannot satisfy a field equation. It is not only ontologically questionable — it is physically inoperable within any known formalism.

.7 Black Holes — Horizon, Singularity, and Fundamental Information

Black Holes Are Points in the Universe, Not the Universe

A black hole is a local, isolated region of the universe — not a model for the universe as a whole, and not a window into a “beyond” of physics. Two relevant horizons exist:

- **The event horizon** — the inner boundary: what lies beyond it is causally disconnected from an external observer.
- **The cosmological horizon** — the outer boundary: what lies beyond our Hubble horizon is not observable to us.

We can only describe what we see. Both horizons are **epistemic** boundaries — not ontological statements about what exists or does not exist beyond them.

Black Holes Are Not Singular in FFGFT

In standard physics, general relativity leads to a singularity inside black holes. In FFGFT, this singularity does not exist [?]:

- $L_0 = \xi \cdot \ell_P$ is an absolute lower bound — infinite density is structurally excluded.
- $\tilde{T} \cdot m = 1$ remains valid even at extreme mass concentrations.
- Black holes are specific T^4 winding configurations — extreme states within the framework’s validity, not exceptions to it.

FFGFT Does Not Lose Fundamental Information

The information paradox [?]: matter falls into a black hole, Hawking radiation is thermal — information appears to be destroyed.

In FFGFT, winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ are *topological invariants* — they cannot disappear. What appears as “information loss” is in FFGFT an **access problem** (epistemic), not an **existence problem** (ontological) [?]. The information is not destroyed — it is inaccessible to external observers behind the horizon.

.8 Self-Referentiality — We Always Describe From the Inside

We Are Part of the System

One might object: our description of FFGFT stands conceptually *above* FFGFT — we occupy a meta-level. This is correct.

But this meta-level is itself only possible because we are materially structured beings. Neurons, electrochemical signals, conscious processes — all are material, all are within T^4 .

The description does not truly stand outside the system. It is a process *within* the system that acts as if it stands above it.

We are part of the system. We always describe from the inside, never from the outside.

The Thought Experiment of an Outside View

The attempt to adopt an outside view through the mind is itself a material process within T^4 . The “mind that stands outside” is a winding configuration on T^4 that *simulates* an external perspective. It does not truly step out.

Every apparent outside view — including on FFGFT itself, including on the question “why T^4 ?” — is an inside view dressed as an outside view.

The Recursive Situation

This gives rise to an unavoidable recursion:

$$\text{Matter} \rightarrow \text{Description} \rightarrow \text{Matter} \rightarrow \text{Description} \rightarrow \dots \quad (3)$$

In FFGFT, this recursion does not terminate — and this is consistent. FFGFT is fractal, open, and recursive (cf. Section 1). The framework always returns to $T^4 + \xi$ — not because this is provable from outside, but because it is the primitive that has been posited *from within*.

Gödel Analogy

This is related to Gödel’s incompleteness theorem [?]: a sufficiently powerful formal system cannot prove its own consistency from its own axioms. FFGFT cannot derive its own foundation from within — which is why “why T^4 ?” remains explicitly open. This is not a gap; it is structural honesty.

Important: [Why T^4 and Not “Pure Information”?] Both posits — T^4 and “pure unrestrained information” — are made from within. But they are not equivalent:

- T^4 is **operational**: it permits derivations (winding numbers, ξ , L_0 , field equations). From $T^4 + \xi$ follow measurable predictions.
- “Pure unrestrained information” is **not operational**: it permits no derivation, no measurement, no distinction. It is a placeholder for what one suspects *before* structure, but it has no internal coherence, because coherence already presupposes structure.

A posit made from within is not automatically legitimate. It must prove itself — in derivations, in connectability, in internal consistency. T^4 proves itself. “Pure information” does not.

Consequence for the Postulate of “Pure Information”

The postulate “pure information before geometry” implicitly assumes an outside view — a standpoint beyond all structure from which pure information becomes visible. But this standpoint is itself a material, structured process within T^4 .

One cannot step outside geometry to see what precedes it — the attempt is already a geometric process. FFGFT accepts this limit and describes consistently from the inside.

.9 The Limit of the First Ground: Philosophical and Theological Parallels

The question “Why T^4 ?” and more fundamentally “Why $\xi = 1.333 \times 10^{-4}$?” is not a weakness of FFGFT. It is the inevitable structure of any theory that is honest. The history of philosophy and theology has reached the same limit from different directions and has each time encountered the same wall — or formulated the same insight, depending on perspective.

Leibniz: Why Is There Something Rather Than Nothing?

Gottfried Wilhelm Leibniz posed the deepest question of metaphysics in 1714: “*Pourquoi il y a quelque chose plutôt que rien?*” — Why is there something rather than nothing? [?] This question is the most general form of the question about the first ground. Leibniz answered with the principle of sufficient reason: everything has a reason. But the reason for the whole cannot come from within the whole — it must lie outside. This led him to God as *necessarium ens* (necessary being) outside the contingent world.

The parallel to FFGFT is direct: “Why ξ and not some other value?” has the same structure as Leibniz’s question — only one level more concrete. The answer FFGFT gives is not Leibniz’s: there is no reason that could be provided from within. The question remains explicitly open — and that is more precise than Leibniz’s recourse to an outside that is not accessible.

Spinoza: No Outside — But Also No Perspective

Baruch de Spinoza resolved the problem radically: *Deus sive Natura* — God or Nature [?]. There is no outside, because God is identical with Nature. This solution avoids the contradiction between inside and outside — but at a price: there is no perspective anymore. Spinoza’s God has no intentions, observes nothing. The question “Why this Nature and not another?” is not answerable in Spinoza’s system, because it would presuppose an outside perspective that does not exist.

FFGFT shares Spinoza’s result: no well-formed outside to T^4 . But FFGFT claims less — it does not say that T^4 is everything, only that T^4 is the framework within which we describe. What lies outside is neither “nothing” nor “God” — it is simply not addressable.

Kant: The Thing-in-Itself Is Unknowable

Immanuel Kant drew a sharper boundary in 1781: between the world as we can know it (phenomena) and the world as it is in itself (noumena, the thing-in-itself) [?]. The thing-in-itself is in principle unknowable — not because we are not intelligent enough, but because cognition is always mediated through the forms of our intuition (space, time) and our understanding (categories). The world as it “really” is, beyond these forms, is not accessible to us.

In FFGFT the parallel holds: what “really” lies outside T^4 is in principle not accessible — not because we lack a sufficiently good theory, but because every description comes from within and thus already presupposes T^4 structure. “Why T^4 ?” is the FFGFT version of Kant’s thing-in-itself — the question about what lies behind the only accessible form of description.

Gödel: No System Proves Its Own Completeness

Kurt Gödel's incompleteness theorem of 1931 showed: in every sufficiently powerful formal system, there are statements that are neither provable nor refutable within the system [?]. A system cannot prove its own consistency from its own axioms.

The analogy to FFGFT is direct and already noted in Dok. 248 §8: "Why ξ ?" is not provable within FFGFT — the answer lies outside the system. This is not a flaw but Gödel's result applied to physical theories: every theory has axioms it cannot ground by itself. FFGFT names this point explicitly, rather than concealing it.

Wittgenstein: Whereof One Cannot Speak

Ludwig Wittgenstein closed the *Tractatus logico-philosophicus* in 1921 with the most famous sentence of analytic philosophy: "*Wovon man nicht sprechen kann, darüber muss man schweigen*" — Whereof one cannot speak, thereof one must be silent [?]. Wittgenstein did not mean mysticism, but a precise boundary: meaningful statements are those that describe states of affairs. What lies outside describability is not sayable — not because it does not exist, but because language (and logic) reaches its limits here.

FFGFT makes the same decision: "Why T^4 ?" is not answerable within the theory. Not because we do not yet know, but because the question lies outside the describable. Leaving explicitly open what one cannot say is a form of Wittgenstein's silence — actively formulated, not passively avoided.

Panentheism: The Contradiction of Inside and Outside

In various theological traditions — especially the panentheism of Alfred North Whitehead [?] and Charles Hartshorne — an attempt is made to resolve the contradiction: God is both in everything (immanent) and beyond everything (transcendent). Everything is from God, with God — and yet he stands outside.

This attempt fails at the same logical structure Johann Pascher identifies here: a system cannot simultaneously fully encompass its own interior *and* stand outside itself. This is not a conceptual error of the theologians — it is the same limit Gödel proved formally. Theological traditions have intuitively recognised this limit and named it "mystery" or "paradox". FFGFT names it geometrically: T^4 has no well-formed outside.

The kabbalistic-Jewish tradition of *Tzimtzum* (Isaac Luria, 16th century) makes the contradiction explicit: God must withdraw ("tzimtzum") to make room for creation [?]. This is a mythological formulation of the same geometric problem: how can the infinite make room for the finite, when there is no outside?

The Common Limit and FFGFT

All these traditions — Leibniz, Spinoza, Kant, Gödel, Wittgenstein, Whitehead, Luria — reach the same limit:

- The first ground cannot ground itself from within itself.
- A complete outside view is either not accessible (Kant), not possible (Spinoza), not sayable (Wittgenstein), or formally excluded (Gödel).
- The attempt to occupy inside and outside simultaneously leads to contradiction (panentheism) or to identification (Spinoza).

FFGFT reaches the same limit from the direction of physics. "Why $\xi = 4/3 \times 10^{-4}$ and not some other value?" belongs to the same category as Leibniz's original question — only one

level more concrete, and therefore more honestly formulable. The answer FFGFT gives is not an answer: it is the explicit acknowledgment of the limit.

[The Question of the First Ground] “Why T^4 ?” and “Why ξ ?” are not questions FFGFT has not yet answered. They are questions that in principle cannot be answered from within — because every answer already presupposes the geometric structure it is supposed to ground. This is Gödel’s incompleteness theorem applied to physical primitives, Kant’s thing-in-itself in geometric formulation, and Wittgenstein’s limit of the sayable in the language of field theory.

FFGFT shares this limit with the deepest traditions of philosophy and theology. The difference: FFGFT names it explicitly, rather than bypassing it through inspiration (theology) or silent positing (many physical theories).

Closing Remark: “In the Beginning Was the Word”

The discussion about the primitive — geometry or information — leads, in principle, consciously or unconsciously, to one of the oldest answers to the same question: “In the beginning was the Word” (John 1:1).

The Logos concept of the Gospel of John is philosophically precise: the Word is structure, order, distinction — exactly what information in the sense of Shannon1948 (Shannon1948) presupposes. The postulate of “pure, unrestrained information” and the Johannine Logos share the same structural problem: both posit a primitive that stands before all structure, and both require someone *outside the system* to speak it.

The Bible resolves this through inspiration — God as an outside perspective made accessible to the human writers. But the writers were human: material beings within the system. The claim of inspiration is itself a claim made from inside about an outside that was never directly accessible.

FFGFT does not take this step. It posits T^4 and ξ as primitive — from within, honestly, without claiming an outside perspective. “Why T^4 ?” remains open. This is structurally more modest than the biblical claim, and more precise.

The difference in one sentence: The Bible claims an outside view it cannot have. FFGFT relinquishes this claim and names the boundary explicitly.

Claim	FFGFT Position
4D is the only possible description	No — 4D is the framework, ontologically open
4D applies to the totality of existence	No — FFGFT makes no total claim
4D applies to the absolutely smallest	Yes — $L_0 = \xi \cdot \ell_P$ sets the lower bound
Below L_0 nothing more exists (in FFGFT)	Yes — no stable degrees of freedom
Geometry is privileged over energy/-mass/time	No — all projections are equivalent
Unstructured information as primitive	Contradiction — information is distinction, hence structure
The vacuum is structureless	No — ξ -field ground state, geometrically structured
A field can exist without structure	No — mathematical tautology

Black holes are singular in FFGFT	No — L_0 structurally prevents infinite density
Black holes destroy information	No — topological invariants are preserved
Horizons are ontological boundaries	No — epistemic limits of the observable
We describe FFGFT from outside	No — we are material T^4 processes
The recursion matter → description terminates	No — open recursion, consistent with FFGFT's character
"Why T^4?" is answered	No — structural honesty, not a gap

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